$$k = \left(\frac{20M}{20min}\right) = 1\,Mmin^{-1}$$

Rate constant for a reaction = $1Mmin^{-1}$

ii) To calculate the time for 80% of completion

$k = 1Mmin^{-1}$, $[A_0] = 100M$, $[A_0]-[A] = 80M$, $t = ?$

Therefore,

$$t = \left(\frac{[A_0]-[A]}{k}\right) = \left(\frac{80M}{1Mmin^{-1}}\right) = 80\,min$$

27 Solution:

Here, we are given that

$E_a = 22.5$ kcal $mol^{-1} = 22500$ cal mol^{-1}

$T = 40°C = 40 + 273 = 313$ K

$k = 1.8 \times 10^{-5}$ sec^{-1}

Substituting the values in the equation

$$\log A = \log k + \left(\frac{E_a}{2.303RT}\right)$$

$$\log A = \log\left(1.8\times10^{-5}\right) + \left(\frac{22500}{2.303\times1.987\times313}\right)$$

$$\log A = \log(1.8) - 5 + (15.7089)$$

$$\log A = 10.9642$$

$$A = \text{antilog}(10.9642)$$

$$A = 9.208\times10^{10} \text{ collisions } s^{-1}$$

28.**Solution:**

For a first order reaction

$$k = \frac{2.303}{t}\log\frac{[A_0]}{[A]}$$

$$k = \frac{2.303}{t}\log\frac{V_\infty}{V_\infty - V_t}$$

In the present case, $V_\infty = 58.3$ ml.

The value of *k* at different time can be calculated as follows:

t (*min*)	V_t	$V_\infty - V_t$	$\frac{2.303}{t}\log\frac{V_\infty}{V_\infty - V_t}$
6	19.3	58.3-19.3 = 39.0	$k = \frac{2.303}{6}\log\left(\frac{58.3}{39}\right) = 0.0670\ min^{-1}$
12	32.6	58.3-32.6 = 25.7	$k = \frac{2.303}{12}\log\left(\frac{58.3}{25.7}\right) = 0.0683\ min^{-1}$
18	41.3	58.3-41.3 = 17.0	$k = \frac{2.303}{18}\log\left(\frac{58.3}{17}\right) = 0.0685\ min^{-1}$
24	46.5	58.3-46.5 = 11.8	$k = \frac{2.303}{24}\log\left(\frac{58.3}{11.8}\right) = 0.0666\ min^{-1}$

Answers.indd 254 22-12-2022 19:40:39